

SENSITIVITY OF EARLY LIFE STAGES OF FRESHWATER MUSSELS (UNIONIDAE) TO ACUTE AND CHRONIC TOXICITY OF LEAD, CADMIUM, AND ZINC IN WATER

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Abstract—Toxicity of lead, cadmium, or zinc to early life stages of freshwater mussels (fatmucket, *Lampsilis siliquoidea*; Neosho mucket, *L. rafinesqueana*) was evaluated in 48-h exposures with mussel larvae (glochidia), in 96-h exposures with newly transformed (5-d-old) and two- or six-month-old juvenile mussels, or in 28-d exposures with two- or four-month-old mussels in reconstituted soft water. The 24-h median effect concentrations (EC50s) for fatmucket glochidia ($>299 \mu\text{g Pb/L}$, $>227 \mu\text{g Cd/L}$, $2,685 \mu\text{g Zn/L}$) and 96-h EC50s for two- or six-month-old fatmucket ($>426 \mu\text{g Pb/L}$, $199 \mu\text{g Cd/L}$, $1,700 \mu\text{g Zn/L}$) were much higher than 96-h EC50s for newly transformed fatmucket (142 and $298 \mu\text{g Pb/L}$, $16 \mu\text{g Cd/L}$, 151 and $175 \mu\text{g Zn/L}$) and Neosho mucket ($188 \mu\text{g Pb/L}$, $20 \mu\text{g Cd/L}$, $145 \mu\text{g Zn/L}$). Chronic values for fatmucket were $10 \mu\text{g Pb/L}$, $6.0 \mu\text{g Cd/L}$, and 63 and $68 \mu\text{g Zn/L}$. When mussel data from the present study and the literature were included in updated databases for deriving U.S. Environmental Protection Agency water quality criteria, mussel genus mean acute values were in the lower percentiles of the sensitivity distribution of all freshwater species for Pb (the 26th percentile), Cd (the 15th to 29th percentile), or Zn (the 12th to 21st percentile). The mussel (*Lampsilis*) genus mean chronic value was the lowest value ever reported for Pb (the 9th percentile) but was near the middle of the sensitivity distribution for Cd (the 61st percentile) or Zn (the 44th percentile). These results indicate that mussels were relatively sensitive to the acute toxicity of these three metals and to the chronic toxicity of Pb, but were moderately sensitive to the chronic toxicity of Cd or Zn compared to other freshwater species. Environ. Toxicol. Chem. 2010;29:2053–2063. © 2010 SETAC

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INTRODUCTION

Freshwater mussels are the most imperiled order of animals in the United States; about 70% of species are listed as endangered, threatened, or of special concern [1,2]. Environmental contamination has been identified as a causal or contributing factor to the decline of mussel populations [2,3]. Natural and human activities, especially activities associated with metal mining and smelting, have resulted in the release of metals into the aquatic environment. Heavy metals can cause mortality, disrupt enzyme efficiency, alter filtration rates, reduce growth, and change behavior of freshwater mussels [4]. The early life stages of mussels are more sensitive to some toxicants, such as Cu and ammonia, compared to commonly tested aquatic organisms [5–7].

Elevated concentrations of some metals, such as Pb, Cd, and Zn, have been reported in water and sediment, potentially creating risks for sensitive invertebrates, fish, and wildlife in some ecosystems [8–10]. However, limited information is available regarding the acute or chronic effects of exposure to Pb, Cd, and Zn on freshwater mussels [4,7]. The present study was conducted in response to concerns regarding mussel exposure to metals in mining-related areas in the Ozark

Plateau region in Arkansas, Kansas, Missouri, and Oklahoma, USA [11].

Most species of freshwater mussels have a complex reproductive cycle that involves a parasitic stage on fish. Fertilized mussel eggs are brooded in specialized chambers of female mussel gills and develop into larvae (called glochidia). Glochidia are released into water and must attach to the gills or fins of a host fish. After one to several weeks of the parasitic stage, glochidia transform to juvenile mussels, detach from the fish, and drop to the stream or lake bottom to begin the free-living juvenile stage. Limited studies indicate that early life stages of freshwater mussels may be more responsive to some toxicants than adult mussels [12].

The objective of the present study was to assess acute or chronic toxicity of Pb, Cd, or Zn in water to glochidia or juveniles of two freshwater mussel species (fatmucket, *Lampsilis siliquoidea*; Neosho mucket, *L. rafinesqueana*). Both species are native to the Ozark Plateau region of the United States. Fatmucket is a common and widespread species in the Midwest, and Neosho mucket is endemic to a small region in the upper Arkansas River system and is a candidate species for federal endangered and threatened species [13]. To evaluate the sensitivity of early life stages of mussels relative to other freshwater species, the toxicity data for mussels in the present study and from previous studies were compared to toxicity data for other freshwater species in the following databases: The Pb database was obtained from Great Lakes Environmental Center (GLEC, Traverse City, MI, USA, 2008 draft update of ambient

All Supplemental Data may be found in the online version of this article.

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aquatic life water quality criteria [WQC]—lead, prepared for the U.S. Environmental Protection Agency [U.S. EPA], Washington, DC, unpublished data); the Cd database was obtained from Mebane ([14]; <http://pubs.water.usgs.gov/sir20065245/>) that was based on the U.S. EPA WQC for cadmium [15]; and the Zn database was obtained from International Zinc Association (IZA, Durham, NC, USA, unpublished data, Eric van Genderen) and an environmental risk assessment in European freshwaters [16].

MATERIALS AND METHODS

Test organisms

Gravid female fatmucket were collected between February and April in 2005 and 2006 from the Silver Fork of Perche Creek (Boone County, MO, USA) and Neosho mucket were collected in July 2005 from Spring River (Cherokee County, KS, USA). Both sites have apparently stable mussel populations with multiple age classes. The adult mussels were transported to the Columbia Environmental Research Center (CERC), U.S. Geological Survey, Columbia, MO, and held in well water (hardness 280 mg/L as CaCO_3 , alkalinity 250 mg/L as CaCO_3 , pH 8.3) at 10°C before isolating glochidia for toxicity testing.

The viability of glochidia isolated from each female mussel was determined before starting a toxicity test [17]. Glochidia were gently flushed from the marsupial gills of a female mussel into a 300-ml crystallizing dish using a 1-mm needle and 35-ml syringe filled with culture water. Three subsamples of about 100 glochidia were impartially transferred to each of three wells of a 24-well polystyrene tissue-culture plate filled with about 2 ml of well water. Glochidia in each well were examined with a dissecting microscope, and the number of closed glochidia was recorded. After adding one drop of saturated NaCl solution to each well, open and closed glochidia were counted within 1 min [17]. Glochidia that closed in response to NaCl were classified as alive (or viable), whereas glochidia that were closed before the addition of NaCl or that remained open after the addition of NaCl were classified as dead (nonviable [17]). Survival (viability) of glochidia was calculated as follows:

$$\text{Survival (\%)} = 100 \times (\text{number of closed glochidia after adding NaCl solution} - \text{number of closed glochidia before adding NaCl solution}) \div (\text{total number of open and closed glochidia after adding NaCl solution}).$$

If the survival of glochidia from an individual mussel was >90%, the remaining glochidia from that mussel were used for toxicity testing. Glochidia isolated from six mussels were pooled and mixed to obtain organisms to start a toxicity test. Glochidia were acclimated to a mixture of 50% culture water (well water) and 50% test water (details follow) that was gradually adjusted to the test temperature over 2 h before the start of a toxicity test.

Field-collected female mussels also were transported to Missouri State University (Springfield, MO, USA) and held in American Society for Testing and Materials (ASTM) reconstituted hard water (160–180 mg hardness as CaCO_3 /L [18]) for propagation of juvenile mussels. Glochidia were recovered and viability was tested as described above. The viability of glochidia from all samples was >80%. Glochidia collected from at least three female mussels were pooled and then placed on host fish for metamorphosis (transformation of juvenile mussels). Host fish were hatchery-reared largemouth bass (*Micropterus salmoides*) of about 8-cm total body length obtained from the Missouri Department of Conservation, Ches-

apeake Fish Hatchery (Chesapeake, MO, USA). The bass were infested with glochidia for 15 min in ASTM reconstituted hard water containing about 4,000 glochidia/L. The infested fish were then transferred to a recirculating aquarium system designed for the recovery of newly transformed juvenile mussels [19]. Water temperature was maintained between 21 and 23°C during the parasitic period. Transformed juveniles left the host fish approximately two to three weeks after inoculation. The newly transformed juveniles derived from a single collection day during the peak drop-off of juveniles from the fish were shipped overnight to the CERC for toxicity testing. Juvenile mussels also were reared in a water-only compact culture system [19] for two to six months with laboratory-cultured algae (*Neochloris oleoabundans*) before shipping to the CERC for toxicity testing.

When juvenile mussels were received at the CERC, the temperature of the holding water was gradually adjusted to the test temperature (<3°C/h) by placing containers of juvenile mussels at room temperature or into a water bath at 20°C. Mussels and shipping water (about 250 ml) were placed into 1-L glass beakers, and a similar amount of temperature-adjusted control water (250 ml) was added into the beaker. About 50% of the water in the beaker was replaced with test water at least three times over the 2- or 3-d acclimation period. Gentle aeration of water was provided to the beaker through a glass pipette (about three air bubbles/second). The mussels were fed 5 ml of a nonviable algal mixture (*Nannochloropsis* and Shellfish Diet, Reed Mariculture) twice daily during the acclimation period (detailed procedures used to prepare the algal mixture are described in Wang et al. [20]).

Acute toxicity test

Test conditions and procedures were in accordance with standard methods [17]. The ASTM reconstituted soft water (hardness 40–48 mg/L as CaCO_3 [18]) was used as test water to represent the lower range of water hardness of the Ozark Plateau region [21]. The reconstituted water was prepared by adding reagent-grade salts ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, MgSO_4 , KCl, and NaHCO_3 ; EM Science) to deionized water [18]. Three toxicants used to prepare the exposure water were lead nitrite ($\text{Pb}[\text{NO}_3]_2$, 99.5% purity), cadmium nitrite ($\text{Cd}[\text{NO}_3]_2$, 99.9% purity) and zinc sulfate (ZnSO_4 , 100% purity). All toxicants were obtained from Sigma-Aldrich. A stock solution of each toxicant was prepared by adding the toxicant to deionized water and was held at 2°C. Each test consisted of five concentrations of Pb, Cd, or Zn in a 50% serial dilution, plus a control in the reconstituted test water. Test solutions were prepared about 1 or 2 h before the start of exposures, except for lead solution. Because of the low solubility of Pb, lead solutions were prepared and held at 20°C for 24 h before the start of the exposures or water renewals.

Glochidia testing. Because limited numbers of gravid Neosho mucket were collected from the field, Neosho mucket glochidia were not used for toxicity tests and only fatmucket glochidia were tested. Each test consisted of three replicates and was conducted under static conditions in 200-ml glass crystallizing dishes (75-mm diameter, 35-mm high) containing about 100 ml of test solution. At the beginning of each test, about 1,000 glochidia were impartially transferred from the pooled sample of glochidia into each replicate chamber. Test chambers were held in temperature-controlled water baths at $20 \pm 1^\circ\text{C}$. Ambient laboratory light of about 200 lux was used with 16:8 h light:dark photoperiod.

Initial survival of glochidia for each toxicity test was estimated by determining the viability of glochidia in three replicate controls at the beginning of the test. A mean viability value was used to adjust the survival of glochidia after 6, 24, or 48 h of exposure (i.e., percent of the initial mean control survival). At each exposure time interval, a subsample of about 100 individuals with 2 ml of test water was taken from each replicate chamber and transferred into one well of a clean 24-well tissue-culture plate. One drop of the saturated NaCl solution was added into the well, and the response of glochidia (valve closure) within 1 min was recorded as described previously. The acceptability criterion for a glochidia test was $\geq 90\%$ control survival at the end of the exposures [17].

Juvenile testing. Water temperature, light quality and intensity, photoperiod, test water, toxicants, solution preparation, and dilution factor in juvenile mussel tests were the same as in glochidia tests. Multiple toxicity tests were performed with newly transformed mussels (about 5-d-old) and with approximately two- and six-month-old juvenile mussels. The duration of juvenile mussel tests was 96 h. Each test consisted of four replicates and was conducted in 50-ml glass beakers containing 30 ml of test solution. At the beginning of each test, five juvenile mussels exhibiting foot movement were impartially transferred into each replicate beaker. Juveniles were not fed during the tests. Test solutions were renewed after 48 h by replacing about 75% of water volume. Survival of juvenile mussels was determined at 48 and 96 h. Juvenile mussels that exhibited foot movement within a 5-min observation period were classified as alive using a dissecting microscope [17]. The acceptability criterion was $\geq 90\%$ control survival at the end of the exposures [17].

Chronic toxicity testing

Two-month-old juvenile fatmucket and Neosho mucket were used to conduct chronic 28-d toxicity tests (Test 1) in 2005. Each species was tested concurrently with Pb, Cd, or Zn in three intermittent flow-through diluter systems [22]. A 28-d zinc test (Test 2) also was conducted starting with four-month-old fatmucket in 2006. Toxicant stock solution was delivered with each cycle of the diluter by a Hamilton syringe pump. Each diluter system delivered five exposure concentrations with a dilution factor of 0.5 plus a control, and provided about 120 ml of test solution to each 300-ml glass beaker every 4 h (about 3.6 volume additions per day). Each beaker had a 2.5-cm hole in the side covered with 50-mesh (279- μm width opening) stainless-steel screen and contained 200 ml of water. Beakers were placed in temperature-controlled water baths at 20°C. At the beginning of the test, 10 juveniles exhibiting foot movement were impartially transferred into each of four replicate beakers per exposure concentration. About 15 juveniles also were sampled impartially and preserved in 70% ethanol to record initial shell lengths. Juveniles were fed 2 ml of the nonviable algal mixture twice daily. The light quality and intensity and photoperiod were as described above for acute toxicity testing.

During the 28-d exposure, survival was determined on test days 4, 10, 21, and 28. Juvenile mussels exhibiting foot movement within a 5-min observation period were classified as alive. The acceptability criterion for a 28-d mussel test was $\geq 80\%$ control survival at the end of the exposures [17]. Surviving juveniles at the end of the tests were preserved in 70% ethanol for subsequent growth measurement. The maximum shell length of each surviving mussel was measured to the nearest

0.001 mm using a digitizing system with video micrometer software (Image Caliper, Resolution Technology).

Reference toxicant testing

To assess the sensitivity of different batches of glochidia or juvenile mussels, a subsample of the batch of test organisms for an acute or chronic toxicity test was evaluated by conducting a 48-h glochidia or 96-h juvenile test in the ASTM reconstituted hard water with six concentrations of copper sulfate (CuSO_4 , 99.9% purity; JT Baker) in a 50% serial dilution. Test methods and conditions were the same as those described above. In addition, recovery tests were conducted in selected reference toxicant (Cu) tests to confirm a previous finding that two-month-old juvenile mussels start to avoid Cu exposure in 96-h toxicity tests by closing their valves [23]. At the end of the selected 96-h test, test organisms were transferred into clean test chambers with control water and held for an additional 24 h at 20°C [17]. Survival (foot movement) of juvenile mussels was determined again after the subsequent 24-h holding period in control water.

Water quality and chemical analysis

Dissolved oxygen, pH, conductivity, hardness, and alkalinity were measured using standard methods [24] on water samples collected from the control, medium, and high exposure concentrations at the beginning and the end of each acute toxicity test, and at the beginning and weekly in each chronic toxicity test. Mean water quality characteristics in test water (Supplemental Data, Tables S1 and S2) were similar to the range of values reported for the ASTM reconstituted soft water (pH 7.2–7.6, hardness 40–48 mg/L, and alkalinity 30–35 mg/L as CaCO_3 [18]) in tests with Pb, Cd, and Zn, and to the range of values reported for the ASTM hard water (pH 7.8–8.0, hardness 160–180 mg/L, and alkalinity 110–120 mg/L as CaCO_3 [18]) in the reference toxicant tests with copper. An exception was higher pH values in the present study, which were consistent with previous studies at the CERC and at other laboratories [20,23]. Mean dissolved oxygen concentration was high (>7.0 mg/L) during each acute and chronic toxicity test.

Water samples (20 ml) for the analysis of Pb, Cd, Zn, or Cu were collected at each exposure concentration at the beginning of 48-h glochidia tests, at the beginning and the end of 96-h juvenile tests, or weekly over the duration of 28-d juvenile tests. Because of limited solubility of Pb, water samples for lead were filtered through a 0.45- μm pore size polypropylene cartridge (Whatman). The water samples for highly soluble Cd, Zn, and Cu were not filtered. All water samples were preserved immediately by adding concentrated nitric acid (16 M, Mallinckrodt Baker) to each sample at a volume proportion of 1:100 (1% v/v).

Concentrations of Cu, Zn, Cd, or Pb were determined by inductively coupled plasma-mass spectrometry ([ICP-MS] PE/SCIEX ELAN DRCE, PerkinElmer). Samples were automatically delivered to the ICP-MS by means of a software-controlled CETAC ASX-500/ADX-100 autosampler/autodiluter system (CETAC Technologies). Internal standards consisted of rhodium (10 $\mu\text{g/L}$), niobium (20 $\mu\text{g/L}$), and iridium (20 $\mu\text{g/L}$). Recoveries of the elements from water reference solutions ranged from 93 to 101%. Analytical precision for quantitative ICP-MS was determined by analyzing samples in duplicate and determining relative percent differences, which ranged from 0 to 7.9%. Recoveries of copper, zinc, cadmium, and lead spiked into water samples ($n = 102$) and analyzed by quantitative ICP-MS ranged from 85 to 107%. Measured concentrations were not adjusted for recovery

efficiency. Method detection limits were calculated for each analytical run, and for each element ranged as follows: 0.03 to 0.21 µg Cu/L, 0.28 to 1.51 µg Zn/L, 0.008 to 0.022 µg Cd/L, and 0.004 to 0.063 µg Pb/L.

Measured concentrations of Cd, Zn, or Cu were similar to nominal concentrations in all acute and chronic tests; mean percentages of nominal concentration ranged from 91 to 115% (Supplemental Data, Tables S1 and S2). As expected, the measured concentrations of Pb were consistently lower than the nominal concentrations due to low solubility of lead in water; the mean percentages of nominal concentration of Pb ranged from 54 to 73% (Supplemental Data, Tables S1 and S2). Measured concentrations of the metals were used for the calculation of acute and chronic effect concentrations of these metals.

Data analysis

Acute and chronic effect concentrations were estimated with TOXSTAT[®] software [25]. Median effect concentrations (EC50s) for survival were calculated using a Probit model [26]. If the data did not meet the requirements of the Probit model, either a Spearman-Kärber or trimmed Spearman-Kärber method was used [26]. No-observed-effect concentration (NOEC) and the lowest-observed-effect concentration (LOEC) for survival and growth at the end of 28-d toxicity tests were determined by analysis of variance (ANOVA) with mean comparison made by one-tailed Dunnett's test, or by Steel's many-one rank test if the data were not normally distributed or did not have equal variances [27]. The level of statistical significance was set at $p \leq 0.05$. Chronic value (ChV) was calculated as the geometric mean of the NOEC and LOEC. Additionally, a linear interpolation method [27] was used to estimate 10% and 20% inhibition concentration (IC10 and IC20) for survival and growth at the end of 28-d toxicity tests.

RESULTS AND DISCUSSION

Acute toxicity

Control survival of glochidia and juvenile mussels was $\geq 90\%$ at the end of all toxicity tests. The EC50s for Cu in reference toxicant tests (Table 1) were within the range of the EC50s obtained from multiple copper toxicity tests in previous studies conducted with fatmucket glochidia and juveniles, or Neosho mucket juveniles [23]. For example, the 24-h EC50 of 33 µg Cu/L for fatmucket glochidia in the present study was comparable to the 24-h EC50s of 29 to 42 µg Cu/L in the previous studies; the 96-h EC50s of 20 to 35 µg Cu/L for juvenile fatmucket in the present study were comparable to 96-h EC50s of 18 to 60 µg Cu/L in the previous studies; and the 96-h EC50s of 23 and 36 µg Cu/L for juvenile Neosho mucket in the present study were comparable to 96-h EC50s of 23 to 43 µg Cu/L in the previous studies. These results from the reference toxicant tests indicate that the test organisms used in the present study had similar sensitivity to Cu toxicity as batches of these organisms tested in the previous studies.

Survival of fatmucket glochidia was over 50% at all six exposure concentrations of Pb, Cd, or Zn over 6, 24, and 48-h exposures (EC50s were greater than the highest exposure concentrations), except for one test where glochidia were exposed to very high concentrations of Zn (24-h EC50 = 2,685 and 48-h EC50 = 1,514 µg Zn/L; Table 1). The EC50s for all 6-h exposures were not included in Table 1 as the 6-h EC50s were all greater than highest exposure concentrations. The 24- and 48-h EC50s for fatmucket glochidia were much

Table 1. Median effect concentrations (EC50s) for survival of glochidia or juveniles of two mussel species (fatmucket, *Lampsilis silicoides*; Neosho mucket, *L. rafinesqueana*) in acute toxicity tests with lead, cadmium, or zinc and in acute copper reference toxicant tests^{a,b}

Test	Species (life stage)	Age	Lead (µg/L)			Cadmium (µg/L)			Zinc (µg/L)			Copper (µg/L, reference test)		
			24 h	48 h	96 h	24 h	48 h	96 h	24 h	48 h	96 h	24 h	48 h	96 h
1	Fatmucket (glochidia)	2 h	>400 ^c	>400 ^c	96 h	>8.0 ^c	>8.0 ^c	>8.0 ^c	>200 ^c	>200 ^c	>200 ^c	NT	NT	24 h after 96-h test ^d
2		2 h	>299	>299		>33	>33	>33	>964	>964	>964	NT	NT	NT
3		2 h	NT	NT		>227	>227	>227	2685 (2599–2775)	1514 (1477–1552)	1514 (1477–1552)	33 (32–34)	26 (25–27)	NT
4	Fatmucket (juveniles)	5 d	465 (427–508)	142 (124–164)	96 h	<34	<34	<34	48 h	48 h	48 h	NT	NT	NT
5		5 d	392 (350–439)	298 (249–357)		42 (39–45)	16 (13–20)	16 (13–20)	192 (183–202)	175 (155–198)	175 (155–198)	32 (26–40)	20 (17–23)	NT
6		2 month	>426	>426		>62	>62	>62	>471	>471	>471	26 (20–33)	35 (27–47)	91 (63–132)
7		6 month	NT	NT		>230	199 (118–337)	199 (118–337)	>2175	1700 (1200–2400)	1700 (1200–2400)	NT	NT	NT
8		2 month ^e	NT	NT		NT	NT	NT	NT	NT	NT	26 (19–36)	31 (25–39)	NT
9	Neosho mucket (juveniles)	5 d	>507	188 (161–220)		38 (34–43)	20 (19–22)	20 (19–22)	134 (115–157)	145 (127–164)	145 (127–164)	57 (47–69)	23 (20–27)	23 (20–27)
10		2 month ^f	NT	NT		NT	NT	NT	NT	NT	NT	39 (34–45)	36 (36–58)	46 (36–58)

^aNumbers in parentheses represent 95% confidence intervals.

^bNT = not tested.

^cEC50s were calculated with nominal concentrations because exposure concentrations were not measured in this test.

^dTest organisms were held in control water for 24 h after the initial 96-h copper exposure.

^eTest was conducted with the same batch of mussels used in a 28-d test with fatmucket (Test 1, Table 2).

^fTest was conducted with the same batch of mussels used in a 28-d test with Neosho mucket (the 28-d test failed due to low control survival on test day 10).

^g95% CI could not be calculated because no partial kills occurred.

higher than 96-h EC50s for 5-d-old juvenile fatmucket or Neosho mucket (Table 1). For example, the 24-h zinc EC50 for fatmucket glochidia (2,685 µg Zn/L) was more than 15-fold higher than the 96-h EC50 for 5-d-old fatmucket juveniles (151 and 175 µg Zn/L).

The EC50s for Pb, Cd, or Zn were similar between newly transformed (5-d-old) fatmucket and Neosho mucket (Table 1). The 96-h EC50s for the newly transformed juvenile fatmucket were much lower than the 96-h EC50s for two- or six-month-old fatmucket (Table 1). For example, 96-h EC50s for six-month-old fatmucket (199 µg Cd/L, 1,700 µg Zn/L) were 9- to 12-fold higher than 96-h EC50 for the newly transformed fatmucket (16 µg Cd/L, 151 and 175 µg Zn/L). Therefore, newly transformed juvenile mussels were more sensitive to the acute toxicity of Pb, Cd, and Zn compared to two- to six-month-old juvenile mussels. These results are consistent with those observed in the reference Cu tests, although the differences in copper EC50s between the ages were within a factor of 2. Wang et al. [23] also found that Cu EC50s for newly transformed mussels were slightly lower than the EC50s for two-month-old mussels.

Three recovery tests were conducted at the end of the reference toxicant test, in which juvenile mussels were held in control water for 24 h after the initial 96-h Cu exposures. The EC50s for newly transformed juveniles did not change after the 24-h holding period in control water, whereas the EC50s for two-month-old juveniles increased substantially (Table 1). These results were similar to findings in a previous study [23], and provide further evidence that two- to four-month-old juvenile mussels may avoid exposure to metals in acute toxicity tests by temporarily closing their valves [23,28]. The ASTM [17] recommends evaluating the 24-h survival of juvenile mussels in control water after 96-h toxicant exposures to determine whether juvenile mussels avoid exposure to a toxicant.

The EC50s for lead and cadmium in juvenile mussel tests decreased substantially with the exposure time from 48 to 96 h, whereas the 48- and 96-h EC50s for zinc were similar (Table 1). The results indicate that zinc was a relatively fast-acting toxicant whereas cadmium and lead were relatively slow-acting toxicants, which is consistent with other freshwater species. For example, cadmium was much slower acting than copper in an acute toxicity test with a cladoceran (*Daphnia pulex* [29]), and than zinc in acute tests with coho salmon (*Oncorhynchus kisutch*) and steelhead trout (*O. mykiss*, formerly *Salmo gairdneri* [30]).

More conservative acute thresholds, such as NOEC and 10% effect concentration (EC10), are useful when a species requires protection from acute exposure [31]. However, the NOEC or EC10 for the metals could not be calculated in the present study because of limited data for analysis of variance to estimate the NOEC, or the data did not fit the Probit model for estimating the EC10. Alternatively, a ratio to predict acute NOECs from acute EC50s was estimated. The NOEC for 5-d-old juvenile mussels in each of three acute exposures (Table 1) was estimated as the highest exposure concentration with ≥90% survival (90% survival was used in the calculation because it is considered acceptable control survival). The geometric mean of the two estimated NOECs for juvenile fatmucket was 112 µg Pb/L, 4.2 µg Cd/L, or 74 µg Zn/L. The geometric mean of the EC50s for juvenile fatmucket was 206 µg Pb/L, 16 µg Cd/L, or 151 µg Zn/L. The ratio of the NOECs to the EC50s was 0.5 for lead, 0.3 for cadmium, or 0.5 for zinc. The estimated NOEC for juvenile Neosho mucket was 58 µg Pb/L, 16 µg Cd/L, or 59 µg

Zn/L. The ratio to the NOEC to the EC50 (Table 1) was 0.3 for lead, 0.8 for cadmium, or 0.4 for zinc. The geometric mean of the ratio of the NOECs to the EC50s for both tested mussel species was 0.4 for Pb, 0.5 for Cd, or 0.4 for Zn, which is consistent with the U.S. EPA guidance of multiplying the Final Acute Value (derived from EC50s for survival) by 0.5 for developing acute water quality criterion [32].

Chronic toxicity

Control survival of juvenile fatmucket ranged from 83 to 100% at the end of 28-d tests, meeting the 28-d test acceptability criterion of ≥80% control survival [17]. However, the 28-d test with juvenile Neosho mucket conducted concurrently in three diluter systems with lead, cadmium, or zinc failed with low control survival (50–63%) on test day 10, although >90% control survival was observed on test day 4 in all three diluter systems. The factors adversely affecting the control survival of Neosho mucket are unknown. The 96-h reference toxicant (Cu) test conducted with the samples from this batch of Neosho mucket in ASTM reconstituted hard water resulted in an EC50 of 36 µg Cu/L (Table 1), which was similar to the 96-h EC50s of 23 and 43 µg Cu/L for juvenile Neosho mucket used in previous Cu tests in the reconstituted hard water [23]. Moreover, a concurrent 28-d sediment test conducted with the same batch of Neosho mucket in diluted well water (hardness 140 mg/L as CaCO₃) exhibited 95% control survival (Chris Ingersoll, unpublished data). These results indicated that the batch of Neosho mucket used in the present study were healthy organisms, and juvenile Neosho mucket survived well in the 28-d sediment exposure with overlying hard water, but may not survive in the reconstituted soft water for more than 10 d. Further study is needed to evaluate the influences of water hardness or substrate on the survival of juvenile Neosho mucket in a long-term exposure. The data from this failed 28-d test with Neosho mucket were not analyzed.

Over 50% mortality of fatmucket was not observed until test day 21 in Pb and Cd exposures, and until test day 28 in zinc exposures (Table 2). The 10-d EC50s for Cd (>22 µg/L) and An (>415 µg/L) in the chronic tests started with two- and four-month-old fatmucket (Table 2) were even higher than the 4-d EC50s for Cd (16 µg/L) and Zn (151 and 175 µg/L) in the acute tests with newly transformed fatmucket (Table 1), further indicating that older juvenile mussels were able to tolerate higher concentrations of Cd or Zn compared to newly transformed juveniles in short-term exposures. However, the EC50 for the three metals in the chronic tests with the older mussels decreased substantially on test days 21 or 28 (Table 2).

Mean length of the two- to four-month-old fatmucket in the controls increased 27 to 49% during the 28-d test period (Table 3) and was similar to the growth of mussels reported in previous 28-d toxicity tests [20]. Chronic value (ChV) for

Table 2. Median effect concentrations (EC50s) for survival of juvenile fatmucket (*Lampsilis siliquoidea*) after 4-, 10-, 21-, and 28-d exposures to lead, cadmium, or zinc in two chronic toxicity tests^a

Test	Toxicant	EC50 (µg/L)			
		4 d	10 d	21 d	28 d
1	Lead	>83	>83	29 (26–33)	20 (17–23)
1	Cadmium	>22	>22	12 (11–13)	8.1 (7.2–9.2)
1	Zinc	>415	>415	>415	333 (271–410)
2	Zinc	>450	>450	>450	228 (202–256)

^a Numbers in parentheses represent 95% confidence intervals.

Table 3. Survival and growth (shell length) of juvenile fatmucket (*Lampsilis siliquoides*) at mean exposure concentrations at the end of 28-d lead, cadmium, or zinc toxicity test^a

Concn. (μg/L)	Lead (Test 1)				Cadmium (Test 1)				Zinc (Test 1)				Zinc (Test 2)			
	Survival (%)	Length (mm)	Concn. (μg/L)	Survival (%)	Length (mm)	Concn. (μg/L)	Survival (%)	Length (mm)	Concn. (μg/L)	Survival (%)	Length (mm)	Concn. (μg/L)	Survival (%)	Length (mm)	Concn. (μg/L)	Survival (%)
0.04	100 (0)	2.15 (0.17)	0.04	100 (0)	2.01 (0.14)	3.1	100 (0)	1.83 (0.20)	2.4	83 (5.0)	1.55 (0.18)					
2.9	100 (0)	2.04 (0.16)	1.2	100 (0)	2.07 (0.13)	21	100 (0)	1.76 (0.17)	24	93 (9.6)	1.36 (0.10)					
6.1	98 (5.0)	1.94 (0.06)	2.2	100 (0)	2.04 (0.09)	45	100 (0)	1.67 (0.12)	48	100 (0)	1.32 (0.13)					
17	78 (25)*	1.63 (0.16)*	4.4	95 (5.8)	1.94 (0.14)	87	100 (0)	1.58 (0.06)*	97	93 (9.6)	1.28 (0.17)*					
36	2.5 (5.0)*	NR ^b	8.2	50 (48)*	NR	179	76 (10)*	1.53 (0.13)*	204	58 (15)*	1.26 (0.09)*					
83	0*	—	22	0*	—	415	40 (32)*	NR	450	0*	—					
NOEC (μg/L)	6.1	6.1		4.4	4.4		87	45		97	48					
LOEC (μg/L)	17	17		8.2	8.2		179	87		204	97					
ChV (μg/L)	10	10		6.0	6.0		125	63		141	68					
IC10 (CI, μg/L)	10 (7.5–17)	6.4 (3.4–11)		4.8 (4.4–8.2)	4.6 (3.8–4.8)		127 (116–148)	55 (24–181)		125 (104–138)	<24					
IC20 (CI, μg/L)	16 (10–19)	14 (9.7–18)		5.7 (5.0–9.7)	5.0 (4.7–5.2)		166 (144–212)	188 (66–208)		154 (132–182)	206 (32–232)					

^a Values are means with standard deviation in parentheses ($n = 4$). An asterisk (*) indicates a significant reduction relative to control ($p < 0.05$). Chronic value (ChV, geometric mean of no-observed-effect concentration [NOEC] and the lowest-observed-effect concentration [LOEC]), 10% and 20% inhibition concentrations (IC10 and IC20 with 95% confidence interval [CI]) are presented for each endpoint. Mean shell length (standard deviation) of juvenile mussels was 1.44 (0.19) mm ($n = 17$) at the beginning of Test 1 and was 1.13 (0.25) mm ($n = 14$) at the beginning of Test 2.

^b NR = not reported due to no survival or only 10% survival of organisms in one or more replicates at this exposure concentration; the length was considered as zero at the concentration when calculating IC10 and IC20.

survival in the Pb or Cd test was identical to ChV for growth, whereas ChVs for growth in the two Zn tests were about half of the ChVs for survival (Table 3). For the same toxicant and endpoint, the ChVs were similar to IC10s and were lower than IC20s, with the exception of Cd, where ChVs were greater than IC10s and IC20s (Table 3). The IC10s and IC20s for growth were typically lower than those for survival. This provides additional evidence that mussel growth is a useful and reproducible sublethal endpoint that should be measured in chronic toxicity tests conducted with juvenile mussels [20].

Chronic toxicity testing is more expensive and time-consuming compared to acute toxicity testing. In addition, specific methods may not be available for conducting chronic toxicity tests with some mussel species. Therefore, acute-chronic ratios (ACRs) are useful for estimating chronic effects from acute toxicity data. The ACRs for Pb, Cd, and Zn were calculated from acute and chronic toxicity data for juvenile fatmucket (i.e., 96-h EC50 for survival divided by 28-d ChV for survival or growth; Table 4). Because newly transformed juvenile mussels were acutely more sensitive to these metals than older juvenile mussels, the 96-h EC50s for 5-d-old juveniles were used to calculate the ACRs (Table 4). The Cd ACR of 2.7 was similar to the final ACR of 3.4 reported in the database for cadmium [14]. The Zn ACR of 2.5 was similar to the final ACR of 2.0 reported in the U.S. EPA WQC for zinc [33], but was about twofold lower than the ACR of 5.3 reported in the updated IZA Zn database. However, the Pb ACR of 22 was much higher than the ACR for Cd or Zn, and was about twofold higher than the final ACR of 9.3 reported in the GLEC Pb database.

Sensitivity of mussels to lead, cadmium, and zinc

The EC50s for Pb, Cd, and Zn in the present study and in previous toxicity tests conducted with glochidia or juvenile mussels are summarized in Table 5. Only historic data that met test acceptability requirements for toxicity test conducted with early life stages of freshwater mussels are included in Table 5. These acceptability requirements included a 24-h duration for glochidia tests and 96-h duration for juvenile mussel tests, in vivo cultured juvenile mussels, measured toxicant concentrations, acceptable water quality, and $\geq 90\%$ control survival [6,17]. No historic Pb toxicity data for freshwater mussels were identified.

Water hardness is used to adjust toxicity estimates for Pb, Cd, or Zn in the current U.S. EPA WQC [15,33] or the updated database ([14]; GLEC lead database, and IZA zinc database). Limited information is available about the influence of water hardness on the toxicity of metals to mussels [7]. Within a limited range of water hardness (40–180 mg/L), the EC50s for Cd or Zn listed in Table 5 were not correlated to water hardness in tests conducted with glochidia ($r^2 < 0.10$, $p > 0.57$), but were significantly correlated to water hardness in tests conducted with juvenile mussels ($r^2 = 0.45$, $p < 0.01$ for Cd; $r^2 = 0.65$,

Table 4. Acute-chronic ratio (ACR) for lead, cadmium, and zinc based on 96-h median effect concentration (EC50) for survival of 5-d-old fatmucket (*Lampsilis siliquoides*) and 28-d chronic values (ChV) for survival or growth of 2- to 4-month-old juvenile fatmucket

Toxicant	EC50 (μg/L)	ChV (μg/L)	ACR
Lead	220 ^a	10	22
Cadmium	16	6.0	2.7
Zinc	163 ^a	66	2.5

^a Mean EC50 of two toxicity tests with 5-d-old fatmucket (see Table 1).

Table 5. Median effect concentration (EC50) for glochidia (24-h exposure) and newly transformed (≤ 5 -d-old) juvenile mussels (96-h exposure) in acute toxicity tests which met the American Society for Testing and Materials (ASTM) test acceptability requirements for toxicity testing with mussels [17]

Chemical	Species	Life stage	Hardness (mg CaCO ₃ /L)	EC50 (Total, μ g/L)	Hardness-normalized EC50 (Total, μ g/L) ^a	Reference
Lead	<i>Lampsilis siliquioidea</i>	Glochidia	47	>332 ^b	>363	This study
		Juveniles	41	154 ^b	205	This study
		Juveniles	47	330 ^b	362	This study
Cadmium	<i>Lampsilis rafinesqueana</i>	Juveniles	42	205 ^b	264	This study
	<i>Lampsilis siliquioidea</i>	Glochidia	48	>33	>34	This study
		Glochidia	46	>227	>239	This study
	<i>Lampsilis cardium</i>	Glochidia	149	>1000	>506	[41]
	<i>Ligumia subrostrata</i>	Glochidia	90	720	499	Clem thesis ^c
	<i>Ptychobranhus occidentalis</i>	Glochidia	90	339	235	Clem thesis
	<i>Utterbackia imbecillis</i>	Glochidia	88	76	53	[42]
		Glochidia	88	50	35	[42]
		Glochidia	88	49	34	[42]
	<i>Lampsilis rafinesqueana</i>	Juveniles	42	20	22	This study
	<i>Lampsilis siliquioidea</i>	Juveniles	42	16	18	This study
	<i>Lampsilis cardium</i>	Juveniles	145	72	37	[43]
	<i>Lasmigona subviridis</i>	Juveniles	86	56	40	[42]
		Juveniles	86	51	36	[42]
		Juveniles	86	67	48	[42]
	<i>Utterbackia imbecillis</i>	Juveniles	83	22	16	[42]
		Juveniles	83	19	14	[42]
		Juveniles	102	50	32	[43]
		Juveniles	102	36	23	[43]
		Juveniles	102	33	21	[43]
Zinc	<i>Actinonaias pectorosa</i>	Glochidia	50	392	392	McCann thesis ^d
		Glochidia	160	1154	431	McCann thesis
	<i>Lampsilis siliquioidea</i>	Glochidia	46	>964	>1035	This study
		Glochidia	48	2685	2779	This study
	<i>Ligumia subrostrata</i>	Glochidia	90	321	195	Clem thesis
	<i>Medionidus conradicus</i>	Glochidia	160	423	158	McCann thesis
		Glochidia	140	549	230	McCann thesis
	<i>Ptychobranhus occidentalis</i>	Glochidia	90	436	265	Clem thesis
	<i>Villosa iris</i>	Glochidia	40	1620	1958	McCann thesis
		Glochidia	50	739	739	McCann thesis
		Glochidia	160	1764	658	McCann thesis
		Glochidia	160	1004	375	McCann thesis
		Glochidia	140	1331	556	McCann thesis
	<i>Actinonaias pectorosa</i>	Juveniles	40	158	190	McCann thesis
		Juveniles	100	292	162	McCann thesis
	<i>Lampsilis rafinesqueana</i>	Juveniles	41	145	172	This study
	<i>Lampsilis siliquioidea</i>	Juveniles	40	175	211	This study
		Juveniles	48	151	156	This study
	<i>Villosa iris</i>	Juveniles	50	209	209	McCann thesis
		Juveniles	50	295	295	McCann thesis
		Juveniles	50	253	253	McCann thesis
		Juveniles	140	516	215	McCann thesis
		Juveniles	140	288	120	McCann thesis
		Juveniles	140	471	197	McCann thesis

^a Values are calculated at a hardness of 50 mg/L based on the equation for lead in the updated lead database (Great Lakes Environmental Center, Traverse City, MI, USA, unpublished data), for cadmium in Mebane [14], and for zinc in the updated zinc database (International Zinc Association, Durham, NC, USA, unpublished data).

^b Total lead value was calculated from dissolved value in Table 1 and a conversion factor (Stephan CE, unpublished report, U.S. Environmental Protection Agency, Duluth, MN, USA).

^c Clem SA, Master's thesis, Arkansas State University, Jonesboro, AR, USA, 1998.

^d McCann MT, Master's thesis, Virginia Polytechnic Institute and State University, Blacksburg, VA, USA, 1993.

$p < 0.01$ for Zn). These results indicate that the acute toxicity of Cd or Zn to juvenile mussels may be influenced by water hardness, similar to the findings with fish or other aquatic invertebrates [15,33]. The hardness-normalized EC50s (hardness 50 mg/L as CaCO₃) for fatmucket or Neosho mucket in the present study were within the range of EC50s reported for mussels in previous studies (Table 5). An exception was the highest Zn EC50 for fatmucket glochidia (2,779 μ g/L) in the present study. The hardness-normalized EC50s for Cd ranged from 14 to 48 μ g/L in juvenile mussel tests (Table 5) and were much lower than the EC50s for Pb (205 to >362 μ g/L) or Zn (120 to 295 μ g/L), which supports

observations that cadmium is more toxic to freshwater mussels than Zn [34,35].

As observed in the present study with fatmucket (Table 1), glochidia tested in previous studies also had higher EC50s for Cd or Zn compared to newly transformed juvenile mussels (Table 5). For example, the hardness-normalized EC50 for Cd was >506 μ g Cd/L for glochidia of plain pocketbook (*Lampsilis cardium*; formerly known as *Lampsilis ventricosa*), but was only 37 μ g Cd/L for juvenile pocketbook. The EC50s for Zn ranged from 375 to 1,958 μ g Zn/L for glochidia of rainbow mussel (*Villosa iris*), but ranged only from 120 to 295 μ g Zn/L for juvenile rainbow mussel.

Furthermore, the overall EC50s for glochidia in all studies were greater than those for juvenile mussels (Table 5). These results indicate that glochidia were less sensitive to the toxicity of the three metals than newly transformed juveniles. This is in contrast to previous findings from acute toxicity tests with Cu, ammonia, and Cl, where glochidia in 24-h exposures and juvenile mussels in 96-h exposures had similar sensitivities to Cu or ammonia, and glochidia were more sensitive to Cl than juvenile mussels [23]. Therefore, there is no consistent pattern in the sensitivity of glochidia relative to the sensitivity of juvenile mussels across all toxicants.

Glochidia have been used routinely to conduct toxicity tests [7,12,17,28]. Testing glochidia may be useful when juvenile mussels are not available due to limited propagation potential such as with an endangered species, or due to the lack of techniques for culturing juveniles of a particular mussel species such as a host fish is unknown [17,28]. However, caution should be used when predicting the acute sensitivity of juveniles from the results of glochidia toxicity tests conducted with some toxicants (as shown here with Cd, Pb, and Zn). The relative sensitivity among life stages of mussels to different toxicants should be further investigated to better understand critical life stages, endpoints, and duration of exposures.

The sensitivity of juvenile mussels to Pb, Cd, and Zn was compared to the sensitivity of freshwater fish and invertebrate species in the updated GLEC lead database, in the Cd database [14], or in the IZA Zn database (acute values) and the Zn database (chronic values) [16]. For the comparison, the 96-h EC50s for newly transformed juvenile mussels from this study and from other studies (Table 5) and 28-d ChVs for two- to four-month-old mussels from this study (Table 3) were first normalized to a hardness of 50 mg/L based on the equations in these updated databases. Genus mean acute values (GMAVs) and genus mean chronic values (GMCVs) for mussels were then calculated as a geometric mean of hardness-normalized EC50s or ChVs of each genus, except that the Pb ChV was not normalized to hardness to be consistent with the approach used in the updated GLEC Pb database. Finally, the GMAVs and GMCVs for all freshwater species in the updated databases were ranked and plotted in cumulative distribution for Pb, Cd, or Zn with addition of the mussel GMAVs and GMCVs (Tables 3 and 5) obtained from the present study and from other studies (Fig. 1). The GMAVs for mussels were in the lower percentiles of the sensitivity distribution for Pb (the 26th percentile, Fig. 1A), Cd (the 15th to 29th percentile, Fig. 1B) or Zn (the 12th to 21st percentile, Fig. 1C). Similarly, the mussel (*Lampsilis*) GMCV for Pb based on the findings from this study was the lowest effect concentration ever reported (the 9th percentile, Fig. 1D). However, the *Lampsilis* GMCV for Cd or Zn was near the middle of the sensitivity distribution (the 61st percentile for Cd, Fig. 1E; and the 44th percentile for Zn, Fig. 1F). These results indicate that mussels were relatively sensitive to acute toxicity of the three metals and to chronic toxicity of Pb, but were moderately sensitive to chronic toxicity of Cd or Zn compared to other tested freshwater species.

The updated GLEC Pb database included no mussel GMAV; the Cd database included four mussel GMAVs [14]; and the updated Zn databases included one mussel GMAV [16; IZA zinc database]. However, the mussel data in these databases are not included in Table 5 because these additional data did not meet the ASTM test acceptability requirements as described above, or because the quality of these additional data could not be evaluated due to lack of information on the test conditions. The sensitivity distribution of GMAVs for those mussels in the

original databases ranged from the 23rd to 31st percentile for Cd and was the 24th percentile for Zn, which were similar to the sensitivity of the mussels listed in Table 5 and plotted in Figures 1 and 2.

Freshwater pulmonate snails of the genus *Lymnaea* are very sensitive to chronic Pb exposure [36]. The Pb GMCV for the mussel *Lampsilis* (10 µg Pb/L) was even lower than the GMCV for the snail *Lymnaea* (18 µg Pb/L; GLEC, unpublished data), becoming the lowest GMCV (Fig. 1D). If the calculation of the Pb GMCVs was based on the ChVs normalized to hardness of 50 mg/L using the equation from the updated database (GLEC, unpublished data), the GMCV for the snails (5.1 µg Pb/L) would be lower than the GMCV for the mussel (9.8 µg Pb/L). Importantly, both the mussels and snails are very sensitive to the chronic toxicity of Pb. In a recent study on physiological mechanisms of the effect of Pb on juvenile pulmonate snail (*L. stagnalis*), Grosell and Brix [37] concluded that the inhibition of Ca^{2+} by Pb effectively controlled chronic toxicity in the snail, likely other freshwater gastropods, and even perhaps mollusks in general. The similar sensitivity of freshwater mussels and snails was observed also in 28-d toxicity tests with ammonia [38]. Additional research is needed to determine why mussels and snails are so sensitive to some metals and ammonia.

More research also is needed to determine why the juvenile mussels are relatively insensitive to chronic exposure of Cd or Zn compared to some other test organisms. Noticeably, the ChVs in the present study were based on 28-d survival and growth of juvenile mussels, which might not be as sensitive to the metals as other chronic test endpoints of other test organisms in the databases, such as reproduction. In addition, because the high mortality of juvenile mussels occurred near the end of 28-d tests (EC50s for the three metals decreased substantially on test days 21 and 28; Table 2), the chronic effect concentrations would likely be lower if longer-term tests were conducted. Future studies are needed to determine the chronic toxicity effect on survival and growth of juvenile mussels in longer exposures (e.g., 90-d exposures) and the effect on reproduction of mussels (e.g., partial life-cycle exposures). In subsequent chronic water-only or sediment toxicity tests with juvenile mussels, we have been photographing shell lengths of mussels in each replicate at the beginning and end of toxicity tests to better account for variance in starting size of mussels, and weighing surviving mussels in each replicate at the end of the toxicity tests to determine the biomass of surviving organisms in a replicate [10,11].

The mussel GMAVs for Pb or Cd were about 2.8- to 55-fold above either the final acute value (FAV) used to derive the respective U.S. EPA acute WQC [39] or the revised FAV in the GLEC Pb database or the Cd database [14] (Fig. 1A and B). However, the mussel GMAVs for Zn were only 1.4- to 1.6-fold above the FAV in the U.S. EPA acute zinc WQC [39] or only 1.2- to 1.5-fold above the FAV in the updated IZA Zn database (Fig. 1C). Thus, including the mussel data may not substantially change the FAV for Pb and Cd, but would likely lower the FAV for Zn, and therefore lower the acute WQC, which is calculated as half of FAV [32]. For chronic data, the mussel GMCV for Pb was about eightfold higher than the U.S. EPA criterion continuous concentration (CCC, i.e., chronic WQC [39]), but was equal to the revised CCC in the updated GLEC database (Fig. 1D; the revised CCC for Pb is about eightfold higher than the current U.S. EPA CCC). The mussel GMCV for Cd was >16-fold higher than both the U.S. EPA CCC [39] and revised CCC [14] (Fig. 1E). The mussel GMCV for Zn was equal to

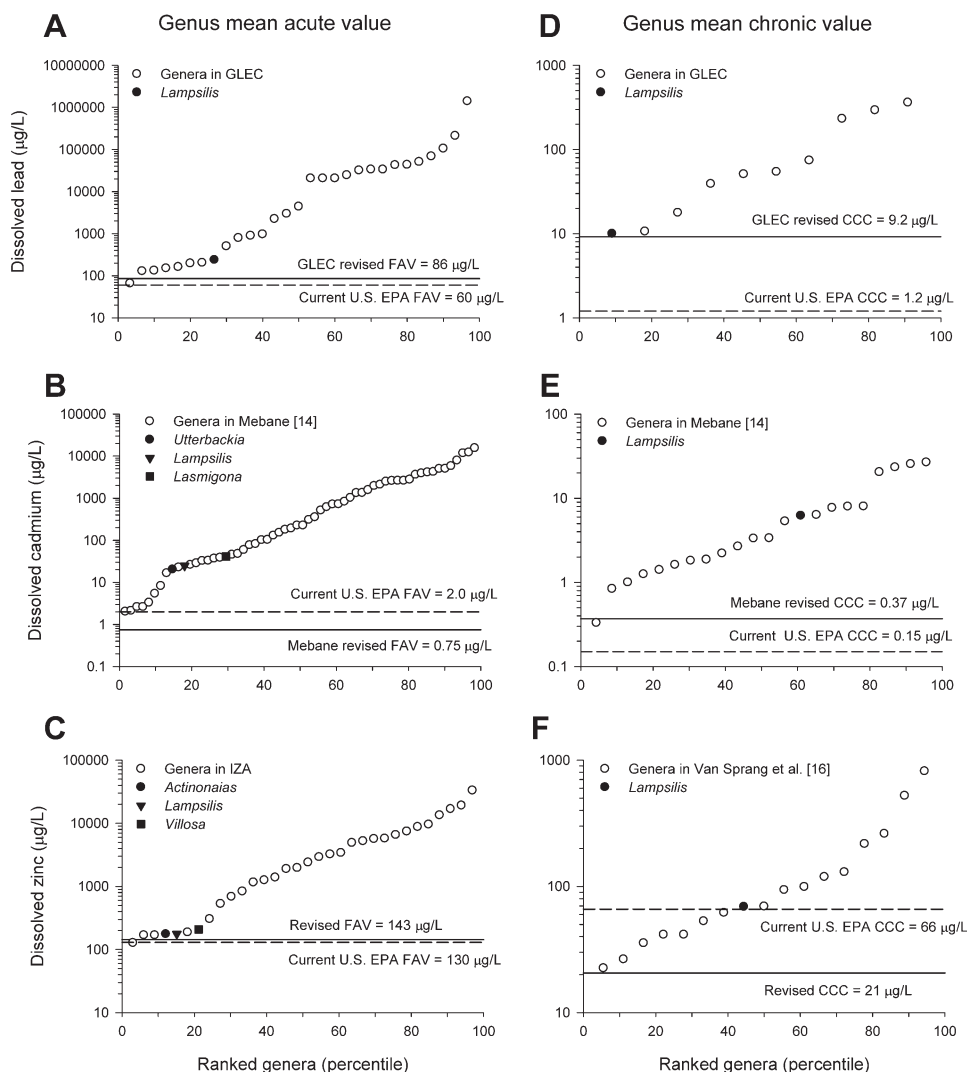


Fig. 1. Ranked freshwater genus mean acute values (GMAVs) (A–C) and genus mean chronic values (GMCVs) (D–F) in the updated database for lead (Great Lakes Environmental Center [GLEC], Traverse City, MI, USA, unpublished data), cadmium [14], or zinc ([16]; International Zinc Association (IZA), Durham, NC, USA, unpublished data) with addition of GMAVs and GMCVs for juvenile freshwater mussels tested in the present study and other studies (Tables 3 and 5). Dashed line indicates the final acute value (FAV) or criterion continuous concentration (CCC) in current U.S. Environmental Protection Agency (U.S. EPA) water quality criteria [39], and solid line indicates the FAV or CCC based on the updated databases. All data and criteria are normalized to a hardness of 50 mg/L as CaCO_3 , except for chronic lead data (Fig. 1D).

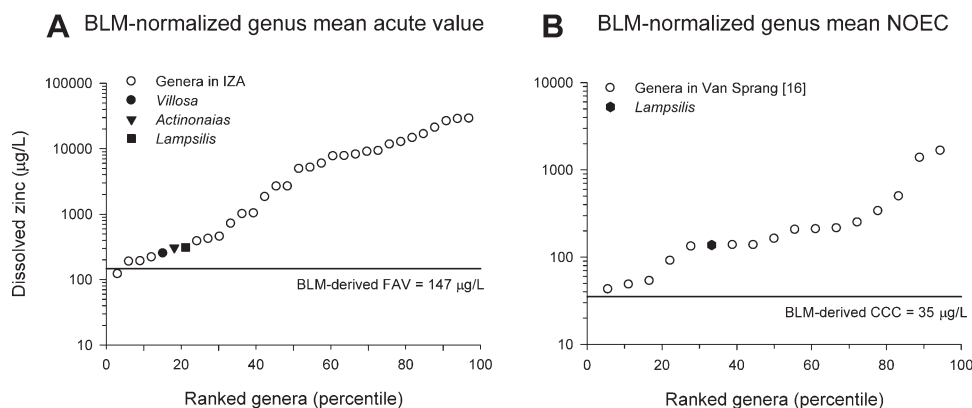


Fig. 2. Ranked freshwater genus mean acute values (GMAVs) (A) and genus mean no-observed-effect concentrations (NOECs) (B) used to derive the draft U.S. Environmental Protection Agency biotic ligand model (BLM)-normalized water quality criterion for zinc ([16]; International Zinc Association (IZA), Durham, NC, USA, unpublished data) with addition of BLM-normalized GMAVs and NOECs for juvenile freshwater mussels tested in the present study and other studies (Tables 3 and 5). Solid line indicates the final acute value (FAV) or criterion continuous concentration (CCC) in the proposed draft BLM-derived water quality criterion ([16]; International Zinc Association (IZA), Durham, NC, USA, unpublished data).

the U.S. EPA CCC, but about threefold higher than the revised CCC for Zn [16] (Fig. 1F; the revised CCC for Zn is about threefold lower than the current U.S. EPA CCC). These results indicate that the current U.S. EPA acute and chronic WQC are protective of the tested freshwater mussels from Pb or Cd exposure, but may not adequately protect the mussels from Zn exposure. The revised acute WQC for the three metals do not substantially change the protection levels for mussels relative to the current U.S. EPA WQC. However, the revised chronic Pb WQC in the GLEC lead database would no longer adequately protect fatmucket, the revised chronic Cd WQC [14] would still be protective of fatmucket, and the revised chronic Zn WQC [16] would improve the protection to fatmucket.

A biotic ligand model (BLM) has been used in the IZA Zn database for a proposed update of the U.S. EPA Zn WQC and in the environmental risk assessment of Zn in European freshwaters [16]. Unlike the current U.S. EPA hardness-dependent WQC for Zn, the proposed BLM-derived WQC criteria are dependent on a number of water quality parameters (e.g., Ca, pH, dissolved organic carbon). To compare the acute and chronic values for other freshwater organisms in the draft WQC, the zinc EC50s and NOECs for juvenile mussels listed in Tables 3 and 5 were normalized using the BLM, which was similar to the approach used to derive the 2007 U.S. EPA WQC for Cu [40]. The BLM-normalized zinc GMAVs for mussels were between the 15th and 21st percentiles in the sensitivity distribution (Fig. 2A), which was similar to the ranks based on hardness-normalized GMAVs (the 12th to 21st percentile; Fig. 1C). The BLM-normalized GMAVs for mussels were still among the most sensitive genera of freshwater species, but were 1.7- to 2.1-fold above the BLM-derived FAV (Fig. 2A). The BLM-normalized Zn NOEC for fatmucket was at the 33rd percentile (Fig. 2B), which was slightly lower than the rank based on the hardness-normalized GMCV (the 44th percentile; Fig. 1F), but was 3.9-fold above the BLM-derived CCC. Therefore, the proposed draft BLM-derived acute and chronic WQC for Zn (Fig. 2) would be more protective of freshwater mussels than the current or revised hardness-dependent WQC for Zn (Figs. 1C and F).

Field studies at metal-contaminated sites indicate that mussel populations may be adversely affected by Zn, Cd, or Pb. Angelo et al. [8] and Besser et al. [10] observed that mussel populations declined with increasing concentrations of Pb, Cd, or Zn in sediment in lead and zinc mining areas in the central United States. Laboratory sediment toxicity testing with juvenile fatmucket exposed for 28 d to sediments collected from these sites also indicated that the survival, growth, or biomass of juvenile mussels were adversely affected by these metals elevated in pore water or in whole sediment [10,11]. Additionally, the sensitivity of the mussels to the metals in these laboratory sediment toxicity tests was similar to, or greater than, two commonly tested benthic organisms (amphipod, *Hyalella azteca*; midge, *Chironomus dilutus*).

CONCLUSION

Newly transformed juvenile mussels (5-d-old) were the most sensitive life stage of freshwater mussels tested in the present study evaluating the acute toxicity of Pb, Cd, or Zn, in contrast to glochidia and older juveniles (two- to six-months-old). In comparison with other tested freshwater organisms, newly transformed juvenile mussels were relatively sensitive to the acute toxicity of Pb, Cd, or Zn, and two- to four-month-old juvenile mussels were relatively sensitive to the chronic toxicity

of Pb, but were moderately sensitive to the chronic toxicity of Cd or Zn. Additional studies are needed to obtain the chronic endpoints comparable to other test organisms, such as survival and growth of juvenile mussels in longer-term exposures (e.g., 90-d exposures) and reproduction of mussels (e.g., partial life-cycle exposures). The current U.S. EPA WQC are protective of freshwater mussels from Pb or Cd exposure, but might not adequately protect the mussels from Zn exposure. The revised hardness-dependent WQC based on updated database for Pb, Cd, or Zn do not substantially change the level of protection for mussels, except that the revised chronic WQC for Pb would no longer be protective of fatmucket, and perhaps other mussel species, and the revised WQC for Zn would be more protective of the mussels. Finally, the proposed draft BLM-derived WQC for Zn would be more protective of mussels compared to the hardness-dependent WQC for Zn.

SUPPLEMENTAL DATA

Table S1. Water quality characteristics and percent nominal concentrations of measured toxicant concentrations in acute toxicity tests with glochidia and juveniles of freshwater mussels (fatmucket, *Lampsilis siliquoidea*; Noesho mucket, *L. rafinesqueana*) in American Society for Testing and Materials (ASTM) reconstituted soft or hard water. (26.6 KB XLS)

Table S2. Mean water quality characteristics ($n = 15$), measured exposure concentrations ($n = 5$), and percent nominal of exposure concentrations ($n = 5$) in chronic 28-d toxicity test with juvenile fatmucket (*Lampsilis siliquoidea*). (20.4 KB XLS)

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